

Thermal radiation effect on MHD stagnation point flow of a Carreau fluid with convective boundary condition

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Abstract: This paper analyzes the effect of thermal radiation on a two-dimensional stagnation-point flow of an in-compressible magneto-hydrodynamic Carreau fluid toward a shrinking surface in the presence of heat transfer and convective boundary condition. Using the similarity transformations, the governing equations have been transformed into a system of ordinary differential equations. The resultant differential equations are solved numerically by using bvp4c MATLAB solver. The behavior of the velocity and temperature as well as skin-friction coefficient and the local Nusselt number for different values of the governing parameters, namely, magnetic parameter, power law index parameter, suction/blowing parameter, radiation parameter and convective parameter are discussed in detail.

Keywords: Carreau fluid, convective surface boundary condition, radiation, Stagnation point flow.

I. INTRODUCTION

Prandtl's boundary layer theory demonstrated to be enormous utilize in Newtonian fluids as the Navier–Stokes equations can be transformed into a great deal basic equations which are easier to handle. The study of boundary layer flow over a stretching sheet is a topic of great attention due to a variety of applications in designing cooling system which included liquid metals, MHD generators, accelerators, pumps and flow meters.

The stretching sheet idea in Crane's (1970) boundary layer flow of a Newtonian fluid past a stretching sheet has been examined and extended by numerous authors. Laminar mixed convection adjacent to vertical, continuously stretching surface studied by Chen (1998).

A stagnation point occurs whenever a flow impinges on a solid object. Pioneering work on 2-dimensional stagnation point flow problem was first studied by Hiemenz who used the similarity transformations approach to reduce the Navier-Stokes equations to non-linear ordinary differential equations. In this contest Nadeem and Hussain (2010) discussed HAM solutions for boundary layer flow in the region of stagnation point towards a stretching sheet. Makinde and Charles (2010) conducted a computational dynamics on the hydrodynamic stagnation point flow towards a stretching sheet.

But little attention has been paid to the four-parameter Carreau inelastic model with the stress formulation

$$\tau_{yx} = \mu_0 \left[1 + \lambda^2 \left(\frac{\partial u}{\partial y} \right)^2 \right]^{\frac{n-1}{2}} \frac{\partial u}{\partial y}$$

is frequently used in chemical engineering. It fits reasonably well with the suspensions of polymers behavior in many flow situations. This model describes the behavior of a purely viscous fluid whose viscosity changes with increasing rate of

deformation. Unlike the power-law or Ostwald-De Waele model, it predicts a viscosity that remains finite as the shear rate approaches zero. For that reason, the Carreau constitutive equation suits well for free surface flows. Among the recent studies in the theory of Carreau fluid include peristaltic flow and heat transfer.

Ali and Hayat (2007) presented the analytic solution of the mathematical modelling for the flow of incompressible Carreau fluid in an asymmetric channel with sinusoidal wall variations. Hayat *et al.* (2010) examined the MHD peristaltic flow of a Carreau fluid in a channel with different waveforms. Olajuwon (2011) investigated the convection heat and mass transfer in a hydromagnetic Carreau fluid past a vertical porous plate in presence of thermal radiation and thermal diffusion. Mohamed Y. Abou-zeid (2009) studied the system of non linear partial differential equations, which describe the unsteady flow of MHD non-Newtonian fluid with heat and mass transfer past a porous plate through a non-Darcy porous medium. Akbar *et al.* (2014) studied the two-dimensional stagnation-point flow of an incompressible Carreau fluid toward a shrinking surface.

Shakhaoath *et al.* (2012) investigated the effect of thermal radiation on non-Newtonian magnetohydrodynamic mixed convective power-Law fluid flow past a vertical stretching sheet with heat generation and chemical reaction effects. Gangadhar (2013) investigated the temperature increases with an increasing the convective parameter. The effects of thermal radiation on MHD stagnation-point flow past a stretching sheet with heat generation was studied by Zhu *et al.*, (2011). El-Arabawy (2003) analyzed the effect of suction and injection on the flow and heat transfer characteristics for a continuous moving plate in a micropolar fluid with the effect of radiation.

The present study investigates the two-dimensional stagnation-point flow of an in-compressible magneto-hydrodynamic Carreau fluid toward a shrinking surface in the presence of heat transfer and convective boundary condition. Using the similarity transformations, the governing equations have been transformed into a set of ordinary differential equations, which are nonlinear and cannot be solved analytically, therefore, bvp4c MATLAB solver has been used for solving it. The results for velocity and temperature functions are carried out for the wide range of important parameters namely; magnetic parameter, stretching/shrinking parameter, convective parameter, radiation parameter and Prandtl number. The skin friction and the rate of heat transfer have also been computed.

II. MATHEMATICAL FORMULATION

Consider a steady laminar two dimensional stagnation point flow of incompressible and electrically conducting Carreau fluid on a stretching surface in the presence of radiation and magnetic field of strength, B_0 , applied in the positive y direction as shown in the figure 1. The induced magnetic field due to the

motion of the electrically conducting fluid and the pressure gradient are neglected.

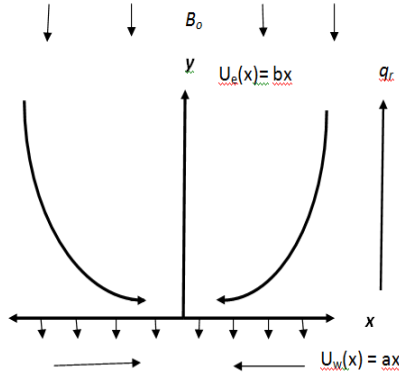


Figure 1. Geometry of the problem

over a wall coinciding with a plane $y=0$, the flow is being confined to $y>0$, the flow is generated due to the linear stretching. Extra stress tensor for carreau fluid is

$$\bar{\tau}_{ij} = \eta_0 \left[1 + \frac{(n-1)}{2} (\Gamma \bar{\gamma})^2 \right] \bar{\gamma}_{ij} \quad (1)$$

In which $\bar{\tau}_{ij}$ is the extra stress tensor, η_0 is the zero shear rate viscosity, Γ is the time constant, n is the power law index and $\bar{\gamma}$ is defined as

$$\bar{\gamma} = \sqrt{\frac{1}{2} \sum_i \sum_j \bar{\gamma}_{ij} \bar{\gamma}_{ji}} = \sqrt{\frac{1}{2} \Pi} \quad (2)$$

Here $\bar{\gamma}$ is the second invariant strain tensor. Flow equations for carreau fluid model and energy equation after applying the boundary layer approximations can be defined as follows.

Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (3)$$

Linear momentum equation

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = u_e \frac{\partial u_e}{\partial x} + n \frac{\partial^2 u}{\partial y^2} + \frac{\sigma B_0^2}{r} (u_e - u) + \frac{3(n-1)G}{2} \frac{\partial}{\partial y} \left(\frac{u \partial^2 u}{\partial y^2} \right) \quad (4)$$

Energy equation

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_1 \frac{\partial^2 T}{\partial y^2} - \frac{\partial q_r}{\partial y} \quad (5)$$

The boundary conditions are

$$u = u_w(x) = ax, v = v_w(x), -\kappa \frac{\partial T}{\partial y} = h[T - T_w] \text{ at } y=0$$

$$u \rightarrow u_w(x) = bx, T \rightarrow T_\infty \text{ as } y \rightarrow \infty \quad (6)$$

where u and v are the velocity components in the x - and y - directions, respectively, T is the fluid temperature in the boundary layer, ν is kinematic viscosity, σ is the electrical conductivity, ρ is the density, and $\alpha_1 (= k / \rho c_p)$ is the thermal diffusivity with k is the fluid thermal conductivity, c_p is the heat capacity pressure, it is noticed that for power law index ($n=1$) our

problem reduced to the case of Newtonian fluid while for $n > 1$ phenomena remains for non-Newtonian fluid respectively. The boundary conditions are

$$u = u_w(x) = ax, v = v_w(x), -\kappa \frac{\partial T}{\partial y} = h[T_w - T] \text{ at } y=0$$

$$u \rightarrow u_w(x) = bx, T \rightarrow T_\infty \text{ as } y \rightarrow \infty \quad (7)$$

In which $b > 0$ is constant, we assume that $u_w(x)=ax$ and $u_e(x)=bx$ are the velocities near and away from the wall respectively, h is the wall heat transfer coefficient. The

radiative heat flux q_r is described by Roseland approximation such that

$$q_r = - \frac{4\sigma^*}{3k^*} \frac{\partial T^4}{\partial y} \quad (8)$$

where σ^* and k^* are the "Stefan-Boltzmann constant" and the "mean absorption coefficient", respectively. We assume that the temperature differences within the flow are sufficiently small so that they T^4 can be expressed as a linear function after using Taylor series to expand T^4 about the free stream temperature T_∞ and neglecting higher-order terms. This result is the following approximation:

$$T^4 \cong 4T_\infty^3 T - 3T_\infty^4 \quad (9)$$

In view of equations (8) and (9), equation (5) reduces to

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_1 \left(1 + \frac{16\sigma^* T_\infty^3}{3kk^*} \right) \frac{\partial^2 T}{\partial y^2} \quad (10)$$

The continuity equation (3) is satisfied by the Cauchy Riemann equations

$$u = \frac{\partial \psi}{\partial y} \text{ and } v = - \frac{\partial \psi}{\partial x} \quad (11)$$

where $\psi(x, y)$ is the stream function.

In order to transform equations (4), (6) and (10) into a set of ordinary differential equations, the following similarity transformations and dimensionless variables are introduced.

$$\eta = \sqrt{\frac{b}{\nu}} y, \psi = \sqrt{b\nu} x f(\eta), \theta = \frac{T - T_\infty}{T_w - T_\infty}, \lambda = \frac{a}{b}, M^2 = \frac{\sigma B_0^2}{\rho b}, Pr = \frac{\nu}{\alpha_1}$$

$$We^2 = \frac{b^3 x^2 \Gamma^2}{\nu}, R = \frac{16\sigma^* T_\infty^3}{3kk^*} \quad (12)$$

where $f(\eta)$ is the dimensionless stream function, θ is the dimensionless temperature, η is the similarity variable, λ is the stretching/shrinking parameter, We is the Weissenberg number, R is the radiation parameter, M is the magnetic parameter, Pr is the Prandtl number.

In view of equations (11) and (12), the equations (4) and (10) transform into

$$f''' + ff'' - (f')^2 + 1 + \frac{3(n-1)We^2}{2} f''' (f')^2 + M^2 (1 - f') = 0 \quad (13)$$

$$\frac{1}{Pr} (1 + R) \theta'' + f \theta' = 0 \quad (14)$$

The corresponding boundary conditions are

$$f(0) = S, f'(0) = \lambda, \theta(0) = -Bi[1 - \theta(0)] \text{ at } \eta=0$$

$$f' = 1, \theta = 0 \text{ as } \eta \rightarrow \infty \quad (15)$$

where the primes denote differentiation with respect to η and the $S = -\frac{v_w}{\sqrt{av}}$, $s > 0$ (i.e. $v_w < 0$) corresponds to suction, $s < 0$

(i.e. $v_w > 0$) corresponds to blowing, $Bi = \frac{h}{\kappa} \sqrt{\frac{\nu}{b}}$ is the convective parameter

The wall shear stress τ_w is given by

$$\tau_w = \frac{\partial u}{\partial x} + \frac{(n-1)\Gamma^2}{2} \left(\frac{\partial u}{\partial y} \right)^3 \quad (16)$$

The coefficient of skin friction is defined as

$$C_f = \frac{\tau_w}{\rho u_w^2}$$

The dimensionless form of skin friction is defined as,

$$\sqrt{\text{Re}_x} C_f = \left[f''(\eta) + \frac{(n-1)We^2}{2} (f''(\eta))^3 \right]_{\eta=0} \quad (17)$$

The dimensionless form of Nusselt number Nu_x is defined as

$$Nu_x = -\frac{x}{T_w - T_\infty} \left(\frac{\partial T}{\partial y} \right)_{y=0} = -\text{Re}_x \theta'(0) \quad (18)$$

Our main aim is to investigate how the values of $f''(0)$ and $-\theta'(0)$ vary in terms of the various parameters.

III. SOLUTION OF THE PROBLEM

The set of equations (13) to (14) were reduced to a system of first-order differential equations and solved using a MATLAB boundary value problem solver called **bvp4c**. This program solves boundary value problems for ordinary differential equations of the form $y' = f(x, y, p)$, $a \leq x \leq b$, by implementing a collocation method subject to general nonlinear, two-point boundary conditions $g(y(a), y(b), p)$. Here p is a vector of unknown parameters. Boundary value problems (BVPs) arise in most diverse forms. Just about any BVP can be formulated for solution with **bvp4c**. The first step is to write the ODEs as a system of first order ordinary differential equations. The details of the solution method are presented in Shampine and Kierzenka[12].

IV. RESULTS AND DISCUSSION

The governing equations (13) - (14) subject to the boundary conditions (15) are integrated as described in section 3. In order to get a clear insight of the physical problem, the velocity and temperature have been discussed by assigning numerical values to the parameters encountered in the problem.

Figure 2 shows the effect of the power law index parameter on the non-dimensional velocity profiles. We observe that the velocity increases with the influence of n in shrinking sheet. These findings are similar to the results reported by Akbar et al., (2014). The results are quite different in the case of stretching sheet.

Figures 3 illustrate the effect of suction/blowing parameter on the velocity for both cases when shrinking case ($\lambda = -3$) and in the absence of shrinking parameter ($\lambda = 0$). We observed that the velocity increases with increasing S . Moreover, the boundary

layer thickness decreases, these results are similar to the findings by Akbar et al. [8].

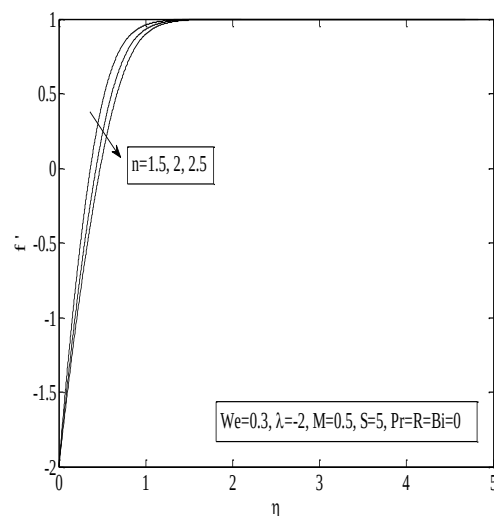


Fig. 2(a) Velocity for different values of n

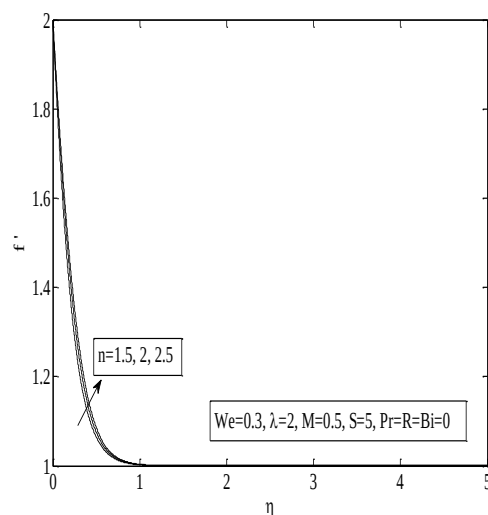


Fig. 2(b) Velocity for different values of n

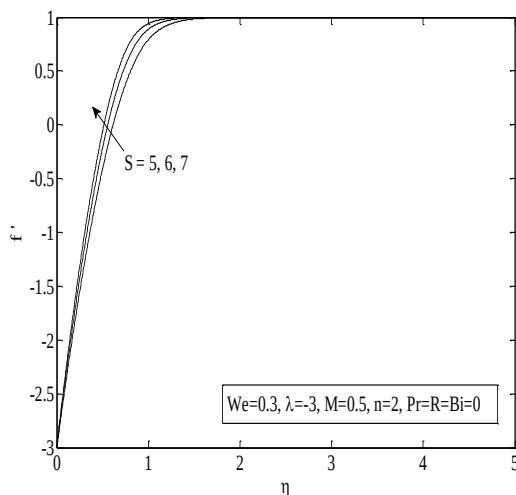


Fig. 3(a). Velocity for different values of S

The variation of the velocity profiles with the magnetic parameter in both cases of shrinking sheet and stretching sheet is shown in Figure 4. It is observed that the velocity increases with an increasing magnetic parameter in case of shrinking sheet and the results are quite different when in the case of stretching sheet. Figures 5, 6 & 7 illustrates the effects of the suction/blowing parameter, convective parameter and radiation parameter on the temperature profiles in both cases of shrinking case ($\lambda = -2$) and stretching case ($\lambda = 2$). It is observed that temperature of the fluid reduces with a rising the suction/blowing parameter and also, it is observed that temperature of the fluid increases with a rising the convective parameter and radiation parameter when both cases of shrinking sheet and stretching sheet.

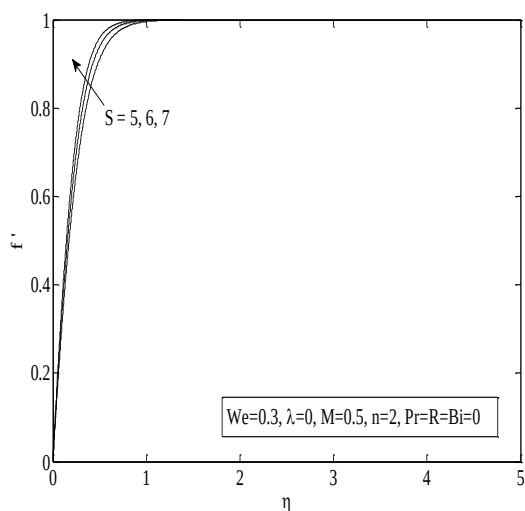


Fig.3(b). Velocity for different values of S

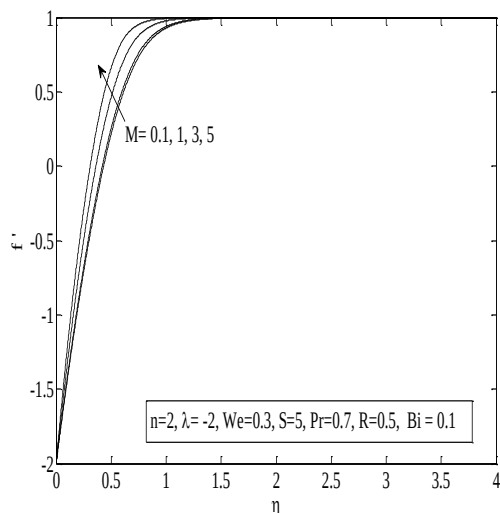


Fig.4(a) Velocity for different values of M

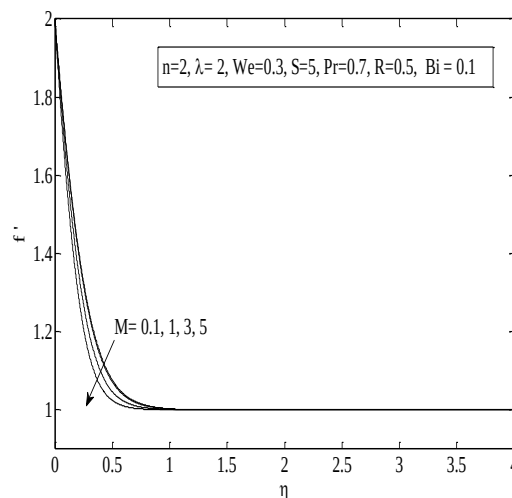


Fig.4(b) Velocity for different values of M

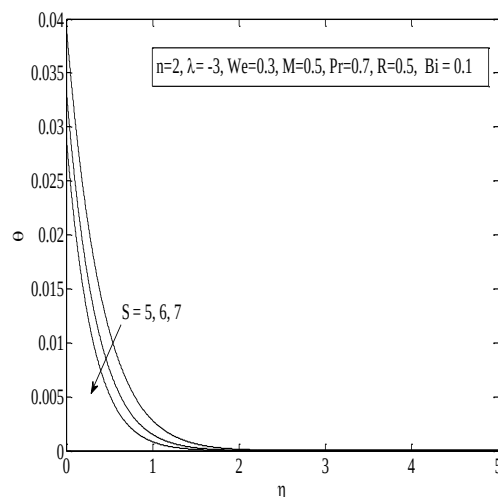


Fig. 5(a) Temperature for different values of S

Variations of the Skin friction and Nusselt number for the different values of S , n , M and λ are presented in Table 1. It is noticed that the skin friction and Nusselt number increases with an increasing the suction/blowing parameter. It is observed that the skin friction and Nusselt number decreases with an increase in the power law index parameter. Also, it is observed that the skin friction reduces where as Nusselt number increases with an increase in the Magnetic parameter or stretching/shrinking parameter.

Table 2 shows that the Nusselt number increases with an increase in the Prandtl number or convective parameter and also observed that Nusselt number reduces with an increasing the radiation parameter.

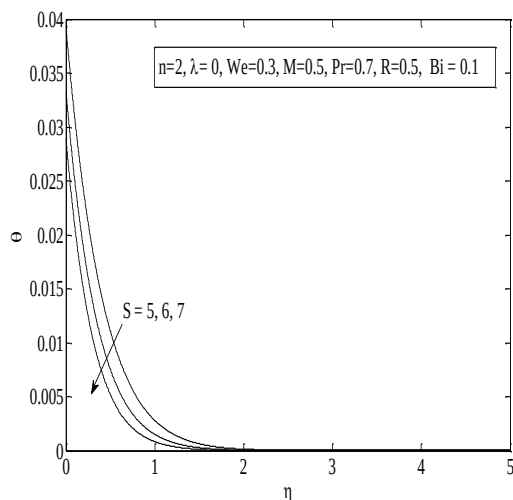


Fig. 5(b) Temperature for different values of S

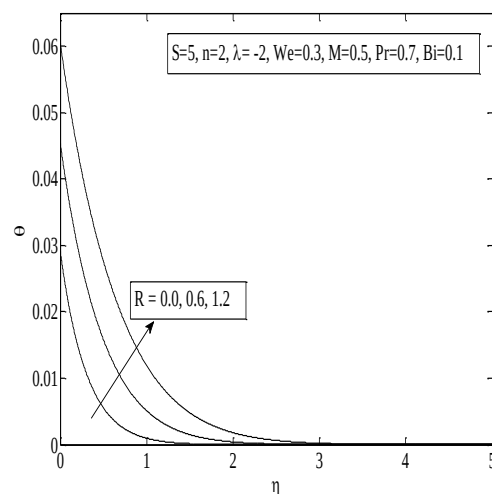


Fig. 7(a) Temperature for different values of R

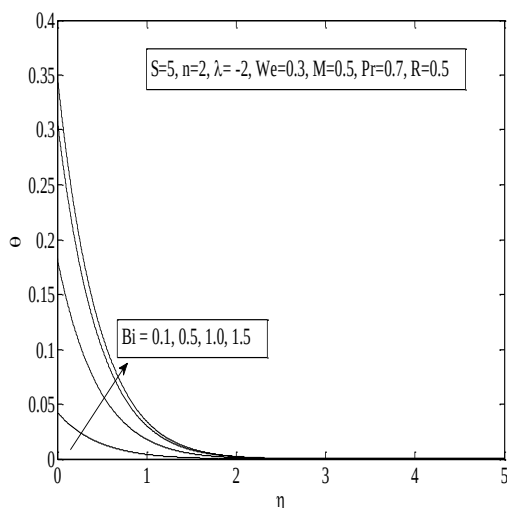


Fig. 6(a) Temperature for different values of Bi

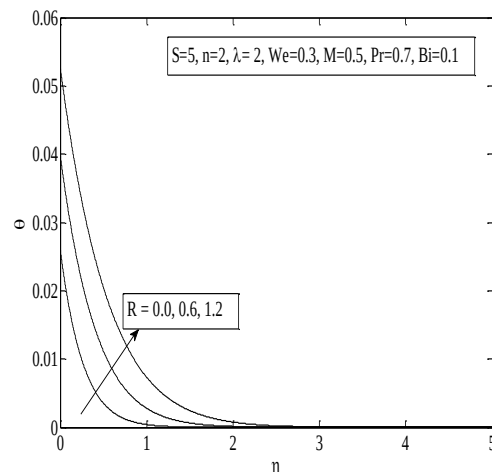


Fig.7(b) Temperature for different values of R

Table.3 shows that the present results perfect agreement to the previously published data.

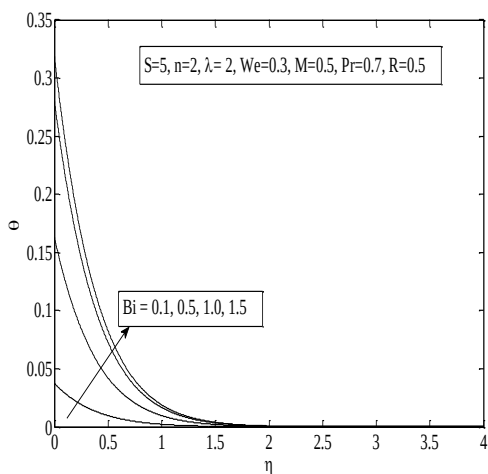


Fig. 6(b) Temperature for different values of Bi

Table.1 Computations for the values of $f''(0)$, $-\theta'(0)$ (with $M=Pr=Bi=We=R=0$) for various values of λ

S	n	m	λ	$f''(0)$	$-\theta'(0)$
5	2	0.5	-3	6.2966	0.0955
6	2	0.5	-3	6.9058	0.0963
7	2	0.5	-3	7.4279	0.0969
5	1.5	0.5	-3	7.7110	0.0955
5	2	0.5	-3	6.2966	0.0955
5	2.5	0.5	-3	5.5618	0.0954
5	2	1	-3	7.1346	0.0955
5	2	2	-3	6.4166	0.0956
5	2	5	-3	5.8295	0.0957
5	2	0.5	-2	3.5186	0.0961
5	2	0.5	0	-3.6780	0.0963
5	2	0.5	2	-6.8117	0.0965

Table.2 Computations for the values of $-\theta'(0)$ (with $M=Pr=Bi=We=R=0$) for various values of λ

Pr	R	Bi	$-\theta'(0)$
3	0.5	0.1	0.0990
5	0.5	0.1	0.0994
7	0.5	0.1	0.0996
0.7	1	0.1	0.0941
0.7	2	0.1	0.0915
0.7	3	0.1	0.0893
0.7	0.5	0.5	0.4039
0.7	0.5	1.0	0.6775
0.7	0.5	1.5	0.8751

Table.3 Comparison for the values of skin friction (with $M=Pr=Bi=We=R=0$) for various values of λ

λ	Present results	Akbar et al. (2014)	Mahapatra and Nandy (2013)	Wang (2008)	Lok et al. (2006)
0	1.2326	1.2326	1.2326	1.2326	---
0.1	1.1466	1.1466	1.1466	1.1466	---
0.2	1.0511	1.0511	1.0511	1.0511	---
0.5	0.7133	0.7133	0.7133	0.7133	0.7133
1	0	0	0	0	---
2	-1.83	-1.887	-1.8873	-1.887	-1.887
5	-10.26	-10.26	-10.264	-10.26	-10.26

III. CONCLUSIONS

In the present paper, a two-dimensional stagnation-point flow of an in-compressible magneto-hydrodynamic Carreau fluid toward a shrinking surface in the presence of heat transfer and convective boundary condition has been studied. The governing equations are approximated to a system of non-linear ordinary differential equations by similarity transformation. Numerical calculations are carried out for various values of the dimensionless parameters of the problem. It has been found that,

1. The velocity decreases in case of Shrinking and increases in the case of stretching with an increase in the power law index parameter.
2. The temperature of the fluid reduces with a rising the suction/blowing parameter and also, increases with a rising the convective parameter and radiation parameter when both cases of shrinking sheet and stretching sheet.
3. The skin friction reduces where as Nusselt number increases with an increase in the Magnetic parameter or stretching/shrinking parameter.

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